INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 2 HIGHER

Expected Paper ELT/U2X Mark Scheme

Step by step, with the marks

M = method mark, A = accuracy mark (needs the method), B = mark for a correct value on its own. Questions 19–26 also show THE IDEA: the one thought that unlocks them.



DR ESLAM AHMED
eliteigcse.com

ANSWERS AT A GLANCE

1	perpendicular bisector constructed
2	$40 < h \leq 60$; 38 cm
3	51.0%
4	\$280
5	6.07×10^{-4}; 4.5×10^{13}
6	$x \geq -3$
7	\$620
8	10
9	\$7462.41
10	10.4 cm; 2.7 cm
11	106°; 148°
12	45; ≈ 15; ≈ 32
13	$F = \frac{180}{d^2}$; 5; 20
14	144π cm^2
15	$k = \frac{2p + 5}{3p - 8}$
16	box plot 18, 24, 31, 39, 52
17	1; $x = -1$; $\frac{4 - x}{x}$
18	$\mathbf{b} - \mathbf{a}$; $\frac{1}{2}(\mathbf{a} + \mathbf{b})$
19	-4; $\left(\frac{2}{3}, \frac{95}{27}\right)$
20	$k = 4$ or $k = -4$
21	7.76 cm
22	$-\frac{1}{6}\mathbf{a} + \frac{1}{2}\mathbf{b}$; $2\mathbf{b}$
23	22
24	$a = 3$, $b = 2$, $c = 1$; 4
25	55°
26	108 cm^2

1 Construction

2 marks

1 **ARCS**
Equal arcs from A and from B (radius more than half of AB), crossing above and below. **M1**

2 **LINE**
Join the two crossing points. **A1**

2 Grouped data

4 marks

3 **(A)**
$40 < h \le 60$ **B1**

4 **(B) MIDPOINTS**
$10 \times 6 + 30 \times 14 + 50 \times 18 + 70 \times 2 = 60 + 420 + 900 + 140 = 1520$ **M1 M1**

ANSWER $1520 \div 40 = 38$ cm **A1**

3 Percentage profit

3 marks

5 **COST AND INCOME**
cost $40 \times 12 = 480$; income $25 \times 20 + 15 \times 15 = 725$ **M1**

6 **PROFIT AS A PERCENTAGE OF COST**
$\frac{725 - 480}{480} \times 100$ **M1**

ANSWER 51.0% **A1**

4 Ratio

3 marks

7 **ONE PART**
Celine − Ali = 5 − 2 = 3 parts = 84, so 1 part = 28 **M1 M1**

ANSWER $10 \times 28 = \$280$ **A1**

5 Standard form

3 marks

ANSWER (a) 6.07×10^{-4} **B1** (b) 4.5×10^{13} **B2**

6 Inequality

3 marks

8 **(A)**

$5 - 20 \le 2x + 3x \Rightarrow -15 \le 5x$ **M1** $x \ge -3$ **A1**

9 **(B)**

solid (filled) circle at −3, arrow to the right **B1**

7 Reverse percentage

3 marks

10 **SALE PRICE IS 70%**

$434 \div 0.7$ **M1 M1**

ANSWER $620 **A1**

8 Polygons

3 marks

11 **INTERIOR + EXTERIOR** = 180

$4e + e = 180 \Rightarrow e = 36^\circ$ **M1 M1**

ANSWER $360 \div 36 = 10$ sides **A1**

9 Compound interest

3 marks

ANSWER $6500 \times 1.028^5 = \$7462.41$ **M1 M1 A1**

10 Similar triangles

4 marks

12 **SCALE FACTOR**

$AB = 8$, so factor $= 8 \div 5 = 1.6$ **M1**

13 **(A)**

$BC = 6.5 \times 1.6 = 10.4$ cm **A1**

14 **(B)**

$AC = 4.5 \times 1.6 = 7.2$ **M1**, so $EC = 7.2 - 4.5 = 2.7$ cm **A1**

11 Circle theorems

4 marks

15 **(A)**

$BCD = 180 - 74 = 106^\circ$ — opposite angles of a cyclic quadrilateral add up to 180° **B1 B1**

16 **(B)**

$BOD = 2 \times 74 = 148^\circ$ — the angle at the centre is twice the angle at the circumference **B1 B1**

12 Cumulative frequency

5 marks

17 **(A) MEDIAN AT 80**

45 minutes **B1**

18 **(B) QUARTILES AT 40 AND 120**

LQ ≈ 37.4, UQ ≈ 52.8 **M1**; IQR ≈ 15 (accept 14–17) **A1**

19 **(C) READ UP FROM 55**

about 128 **M1**; 160 − 128 = 32 (accept 30–34) **A1**

13 Inverse proportion

4 marks

20 **(A)**

$F = \dfrac{k}{d^2}$: $20 = \dfrac{k}{9} \Rightarrow k = 180$ **M1 A1**

ANSWER (a) $F = \dfrac{180}{d^2}$ (b) $\dfrac{180}{36} = 5$ **B1** (c) $d^2 = 400$, $d = 20$ **B1**

14 Sphere

3 marks

21 **RADIUS**

$\frac{4}{3}\pi r^3 = 288\pi \Rightarrow r^3 = 216 \Rightarrow r = 6$ **M1 A1**

ANSWER $4\pi(6^2) = 144\pi$ cm^2 **A1**

15 Change the subject

4 marks

22 **MULTIPLY OUT**

$3pk - 2p = 8k + 5$ **M1**

23 ***k* TERMS TOGETHER**

$3pk - 8k = 2p + 5$ **M1**

24 **FACTORISE**

$k(3p - 8) = 2p + 5$ **M1**

ANSWER $k = \dfrac{2p + 5}{3p - 8}$ **A1**

16 Box plot

3 marks

25 **UPPER QUARTILE**

$24 + 15 = 39$ **B1**

ANSWER box from 24 to 39 with the median line at 31; whiskers to 18 and 52 **B2**

17 Functions

4 marks

26 **(A)**

$g(3) = 1$, $f(1) = 1$ **B1**

27 **(B)**

$x = -1$ (the denominator would be 0) **B1**

28 **(C)**

$y = \dfrac{4}{x + 1} \Rightarrow x + 1 = \dfrac{4}{y} \Rightarrow x = \dfrac{4}{y} - 1$ **M1**

ANSWER $g^{-1}(x) = \dfrac{4 - x}{x}$ **A1**

18 Vectors

3 marks

ANSWER (a) $\mathbf{b} - \mathbf{a}$ **B1** (b) $\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{b})$ **M1 A1**

19 Gradient of a curve

5 marks

THE IDEA "the same gradient" means set $\frac{dy}{dx}$ equal to the gradient at A — it is a quadratic, and you already know one root.

29 **DIFFERENTIATE**
$\frac{dy}{dx} = 3x^2 - 8x$; at $x = 2$: $12 - 16 = -4$ **M1 A1**

30 **SAME GRADIENT**
$3x^2 - 8x = -4 \Rightarrow 3x^2 - 8x + 4 = 0 \Rightarrow (3x - 2)(x - 2) = 0$ **M1**

31 **THE OTHER ROOT**
$x = \frac{2}{3}$; $y = \frac{8}{27} - \frac{16}{9} + 5 = \frac{95}{27}$ **M1**

ANSWER $\left(\frac{2}{3}, \frac{95}{27}\right)$ **A1**

20 Line tangent to a circle

5 marks

THE IDEA a tangent meets the circle at ONE point, so the quadratic has one repeated root: $b^2 - 4ac = 0$.

32 **SUBSTITUTE**
$x^2 + (x + k)^2 = 8 \Rightarrow 2x^2 + 2kx + k^2 - 8 = 0$ **M1 A1**

33 **ONE ROOT**
$(2k)^2 - 4(2)(k^2 - 8) = 0$ **M1**

34 **SOLVE**
$4k^2 - 8k^2 + 64 = 0 \Rightarrow k^2 = 16$ **M1**

ANSWER $k = 4$ or $k = -4$ **A1**

21 Frustum into a sphere

5 marks

THE IDEA the small cone is similar to the big one with length factor $\frac{1}{3}$, so its volume is $\frac{1}{27}$ of the big cone.

35 **BIG CONE**
$\frac{1}{3}\pi(9^2)(24) = 648\pi$ **M1**

36 **SMALL CONE**
radius 3 (or $648\pi \div 27$): 24π **M1**

37 **FRUSTUM**
$648\pi - 24\pi = 624\pi$ **A1**

38 **SPHERE**
$\frac{4}{3}\pi r^3 = 624\pi \Rightarrow r^3 = 468$ **M1**

ANSWER $r = 7.76$ cm **A1**

22 Vectors — where two lines meet

5 marks

THE IDEA R is on line PQ AND on line OB: write $\vec{OR}$ both ways, then make the **a** part vanish.

39 **(A)**
$\vec{PQ} = \vec{PA} + \vec{AQ} = \frac{1}{3}\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = -\frac{1}{6}\mathbf{a} + \frac{1}{2}\mathbf{b}$ **M1 A1**

40 **(B) ON LINE PQ**
$\vec{OR} = \frac{2}{3}\mathbf{a} + t\left(-\frac{1}{6}\mathbf{a} + \frac{1}{2}\mathbf{b}\right)$ **M1**

41 **ON LINE OB: NO a**
$\frac{2}{3} - \frac{t}{6} = 0 \Rightarrow t = 4$ **M1**

ANSWER $\vec{OR} = 2\mathbf{b}$ **A1**

23 Arithmetic series — an inequality

4 marks

THE IDEA write the sum in terms of n, set it > 1000, solve the quadratic, and take the next whole number.

42 **SUM**

$S_n = \frac{n}{2}[14 + 4(n - 1)] = n(2n + 5)$ **M1**

43 **INEQUALITY**

$2n^2 + 5n - 1000 > 0$; root $n = \dfrac{-5 + \sqrt{8025}}{4} = 21.1\ldots$ **M1**

44 **CHECK**

$S_{21} = 987$, $S_{22} = 1078$ **M1**

ANSWER 22 terms **A1**

24 Trigonometric graph

4 marks

THE IDEA the middle line gives c, the distance from the middle to the top gives a, the number of waves in 360° gives b.

45 **(A)**

max 4, min −2: $c = 1$, $a = 3$; two waves in 360°: $b = 2$ **B3**

46 **(B)**

the line $y = 2.5$ crosses the curve 4 times **B1**

ANSWER $a = 3$, $b = 2$, $c = 1$; 4 solutions

25 Circle theorems with algebra

5 marks

THE IDEA the cyclic quadrilateral gives an equation for x; the alternate segment theorem moves the tangent angle inside the triangle.

47 **CYCLIC QUADRILATERAL**

$(3x + 10) + (2x + 20) = 180$ — opposite angles of a cyclic quadrilateral **M1**

48 **SOLVE**

$5x = 150 \Rightarrow x = 30$, so $ADC = 80°$, $DCS = 45°$ **A1**

49 **ALTERNATE SEGMENT**

$CAD = DCS = 45°$ — alternate segment theorem **M1**

50 **TRIANGLE ACD**

$ACD = 180 - 80 - 45$ — angles in a triangle **M1**

ANSWER 55° **A1**

26 Minimum surface area

6 marks

THE IDEA use the volume to write the height in terms of x, so the area has only one letter — then differentiate.

51 **HEIGHT**

$2x \cdot x \cdot h = 72 \Rightarrow h = \dfrac{36}{x^2}$ **M1**

52 **AREA**

$A = 2(2x \cdot x) + 2(2x \cdot h) + 2(x \cdot h) = 4x^2 + 6xh$ **M1**

53 **SUBSTITUTE**

$A = 4x^2 + 6x \cdot \dfrac{36}{x^2} = 4x^2 + \dfrac{216}{x}$ **A1**

54 **DIFFERENTIATE**

$\dfrac{dA}{dx} = 8x - \dfrac{216}{x^2} = 0$ **M1**

55 **SOLVE**

$x^3 = 27 \Rightarrow x = 3$ **M1**

ANSWER $A = 36 + 72 = 108\ \text{cm}^2$ **A1**